

	Reg. No.:	23BAI10917
	Name :	Vijay Rajesh R



VIT[®]
B H O P A L
www.vitbhopal.ac.in

TERM END EXAMINATIONS (TEE) – April 2026

Programme	: B.Tech.	Semester	: Winter Semester 2025-26
Course Name	: DATA MINING AND DATA WAREHOUSING	Course Code	: CSA3006
Date/ Session	: 13-4-2026 / Session-I	Slot	: B11+B12+B13
Time	: 3 Hrs.	Max. Marks	: 100

Answer ALL the Questions

Q. No.	Question Description	Marks
--------	----------------------	-------

PART A – (60 Marks)

1	(a)	A retail company wants to analyze its business performance, including sales and shipping operations. The following data is available: - Sales details: Sales_ID, Date, Product_ID, Customer_ID, Quantity, Sales_Amount - Shipping details: Shipping_ID, Date, Product_ID, Customer_ID, Shipping_Cost - Product details: Product_ID, Product_Name, Category, Brand - Customer details: Customer_ID, Customer_Name, City, State - Time details: Date, Day, Month, Year Draw the following diagrams clearly: Star Schema, Snowflake Schema and Fact Constellation Schema	12
OR			
	(b)	Explain data mining as an essential step in the knowledge discovery in databases process. Illustrate the complete process with a suitable block diagram.	12

2	(a)	The following table shows the transaction data of items purchased in a supermarket. 1 = Purchased, 0 = Not Purchased <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <th>TID</th> <th>Milk</th> <th>Bread</th> <th>Butter</th> <th>Eggs</th> </tr> <tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td></tr> <tr><td>2</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>3</td><td>0</td><td>1</td><td>1</td><td>0</td></tr> <tr><td>4</td><td>1</td><td>0</td><td>1</td><td>1</td></tr> <tr><td>5</td><td>1</td><td>1</td><td>1</td><td>0</td></tr> <tr><td>6</td><td>0</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>7</td><td>1</td><td>0</td><td>0</td><td>1</td></tr> </table> Consider the following association rules: Milk, Bread → Butter Milk, Eggs → Bread Bread, Butter → Eggs (i) For each rule, compute the support, confidence, lift, leverage and conviction. (ii) Whether the antecedent increases the likelihood of the consequent. (iii) Identify the rule with the strongest association.	TID	Milk	Bread	Butter	Eggs	1	1	1	0	1	2	1	1	1	1	3	0	1	1	0	4	1	0	1	1	5	1	1	1	0	6	0	1	1	1	7	1	0	0	1	12
TID	Milk	Bread	Butter	Eggs																																							
1	1	1	0	1																																							
2	1	1	1	1																																							
3	0	1	1	0																																							
4	1	0	1	1																																							
5	1	1	1	0																																							
6	0	1	1	1																																							
7	1	0	0	1																																							

	OR																																															
	(b)	Explain the concepts of data objects and attribute types in data mining and data analysis. Discuss the role of data objects in representing real-world entities and how attributes define their characteristics. Provide a comparative analysis of the different attribute types.	12																																													
3	(a)	Consider the following dataset of 10 students. The target variable is Result (Pass/Fail). Identify the root node of the decision tree and construct the final decision tree. <table><tr><th>Student</th><th>Study Hours</th><th>Attendance</th><th>Result</th></tr><tr><td>S1</td><td>Medium</td><td>Poor</td><td>Fail</td></tr><tr><td>S2</td><td>High</td><td>Good</td><td>Pass</td></tr><tr><td>S3</td><td>Low</td><td>Average</td><td>Fail</td></tr><tr><td>S4</td><td>Medium</td><td>Good</td><td>Pass</td></tr><tr><td>S5</td><td>High</td><td>Poor</td><td>Pass</td></tr><tr><td>S6</td><td>Low</td><td>Good</td><td>Fail</td></tr><tr><td>S7</td><td>Medium</td><td>Average</td><td>Pass</td></tr><tr><td>S8</td><td>High</td><td>Good</td><td>Pass</td></tr><tr><td>S9</td><td>Low</td><td>Poor</td><td>Fail</td></tr><tr><td>S10</td><td>Medium</td><td>Average</td><td>Fail</td></tr></table>	Student	Study Hours	Attendance	Result	S1	Medium	Poor	Fail	S2	High	Good	Pass	S3	Low	Average	Fail	S4	Medium	Good	Pass	S5	High	Poor	Pass	S6	Low	Good	Fail	S7	Medium	Average	Pass	S8	High	Good	Pass	S9	Low	Poor	Fail	S10	Medium	Average	Fail	12	
Student	Study Hours	Attendance	Result																																													
S1	Medium	Poor	Fail																																													
S2	High	Good	Pass																																													
S3	Low	Average	Fail																																													
S4	Medium	Good	Pass																																													
S5	High	Poor	Pass																																													
S6	Low	Good	Fail																																													
S7	Medium	Average	Pass																																													
S8	High	Good	Pass																																													
S9	Low	Poor	Fail																																													
S10	Medium	Average	Fail																																													
	OR																																															
	(b)	Discuss the different types of regression analysis by explaining each model with suitable examples, describing their key characteristics and applications, comparing the models based on their features, and highlighting the differences between each regression technique.	12																																													
4	(a)	Apply the K-Means clustering algorithm with $K = 2$ to the dataset: (595,482), (680,366), (478,370), (289,378), (492,282), (298,587), (365,498), (694,888). Perform clustering for three iterations by initially selecting the first two data points as centroids. Use Euclidean distance for cluster assignment, update the centroids after each iteration, and present the final clusters obtained.	12																																													
	OR																																															
	(b)	Describe the following clustering methods. a. Fuzzy Clustering b. Statistical-Based c. Graph-Based Methods Explain their working principles in detail. Illustrate each method with suitable diagrams. Highlight the important terminology associated with each technique. Provide a clear understanding of how these clustering approaches differ in practice.	12																																													
5	(a)	Explain the different types of ensemble methods in detail, describing their architecture with suitable diagrams and discussing the working principles of each approach, while also highlighting the key features and advantages of ensemble techniques, and providing a comprehensive comparison between the various ensemble methods based on their performance, complexity and application areas.	12																																													
	OR																																															
	(b)	(i) Explain in detail the process of data mining on different kinds of data, with suitable real-world examples for each type. Discuss how data mining techniques are applied across these data forms and highlight the challenges involved in handling them. (ii) Provide a comprehensive comparison between structured and unstructured data.	12																																													

PART B – (40 Marks)

6	Construct a table that clearly highlights at least eight distinctions between OLTP and OLAP systems.	8														
7	Consider the following transaction dataset: <table><tr><th>Transaction ID</th><th>Items</th></tr><tr><td>T1</td><td>Pizza, Coke, Fries</td></tr><tr><td>T2</td><td>Pizza, Coke</td></tr><tr><td>T3</td><td>Burger, Coke, Fries</td></tr><tr><td>T4</td><td>Burger, Fries</td></tr><tr><td>T5</td><td>Fries, Juice</td></tr><tr><td>T6</td><td>Pizza, Coke, Fries</td></tr></table> Find the frequent itemsets and generate association rules for the above dataset using Apriori algorithm. Assume that Minimum Support Threshold (s) = 33% and Minimum Confidence Threshold (c) = 70%.	Transaction ID	Items	T1	Pizza, Coke, Fries	T2	Pizza, Coke	T3	Burger, Coke, Fries	T4	Burger, Fries	T5	Fries, Juice	T6	Pizza, Coke, Fries	8
Transaction ID	Items															
T1	Pizza, Coke, Fries															
T2	Pizza, Coke															
T3	Burger, Coke, Fries															
T4	Burger, Fries															
T5	Fries, Juice															
T6	Pizza, Coke, Fries															
8	A team is building a classification model to predict whether a patient has a disease. However, the model is not giving good accuracy. To improve performance, the team tries different methods such as data preprocessing, selecting important features, and using different algorithms. They also evaluate the models using techniques like accuracy, precision, recall, and cross-validation. Finally, they compare different models to choose the best one for prediction. (i) How can the team improve classification accuracy? (ii) What methods can be used to evaluate the model? (iii)How can the best model be selected?	8														
9	Describe the various types of data in cluster analysis, explain their characteristics and support your answer with appropriate examples.	8														
10	A bank maintains records of customer transactions, including account details, transaction amounts, and timestamps. However, the dataset contains missing values and anomalies, which may affect the accuracy of analysis. The bank wants to clean and preprocess the data, and then apply suitable data mining techniques to identify fraudulent transactions and build a reliable prediction model. (i) What preprocessing steps should be applied to clean the dataset? (ii) How can missing values and anomalies be handled? (iii)Which data mining techniques can be used to detect fraudulent transactions? (iv)How can a reliable prediction model be built from this data? (v) What challenges might arise during fraud detection?	8														

